This is a group assignment for the project. Each person in the group should make their own submission of the group work.

Project proposal should be around 2 pages.

1. Explain the problem you are trying to solve in this project.
2. Explain your approach and technical design plan to implement the project.
3. Explain the cybersecurity concepts the project will cover.
4. Explain the ethical issues that will be explored as they connect to the project problem, design and cybersecurity concepts.
5. Attribute group member authorship for the above proposal components.

Rubric:

- Problem, approach and technical design plan - up to 12 points
- Cybersecurity and ethical issues - up to 12 points
- Clarity and organization of proposal - up to 6 points

Project Proposal

Project Title: Autonomous Phishing Email Detector & Responder
Course: CSCI 401
Team Name: John Jay Seniors

1. Problem Statement

Phishing attacks remain among the most common cybersecurity threats today. Attackers often send emails that appear legitimate to trick users into revealing sensitive information, such as passwords, financial details, or personal data. These emails often impersonate trusted organizations such as banks, delivery companies, or workplace systems. Because many phishing emails are carefully crafted, it can be difficult for users to recognize them.

Organizations receive large volumes of emails every day, making it difficult for security teams to manually review each message. Even with spam filters and basic security tools, many phishing attempts still reach users. When a phishing email is not detected in time, it can lead to account compromise, data breaches, or financial loss.

The goal of this project is to develop an **Autonomous Phishing Email Detector & Responder**, a system that can automatically analyze emails and identify potential phishing attacks. The system will use artificial intelligence and cybersecurity analysis to detect suspicious patterns and indicators commonly found in phishing emails. Once a suspicious email is detected, the system will generate warnings or recommended responses to help users or administrators handle the threat.

By building this system, the project aims to demonstrate how automation and AI can assist in improving email security and reducing the risk of phishing attacks.

2. Approach and Technical Design Plan

Our project will be designed as a system that combines artificial intelligence, data processing, and cybersecurity analysis. The goal is to automatically analyze email content and determine whether it is legitimate or potentially malicious.

The first step will involve collecting and preparing a dataset of phishing emails and legitimate emails. This data will be cleaned and organized so that it can be used for analysis and testing. The dataset will help the system learn patterns commonly found in phishing attacks.

The main component of the system will be an **AI-based phishing detection model**. Using natural language processing and machine learning techniques, the system will analyze email content and identify suspicious characteristics. For example, the model may detect signs such as urgent language, requests for login credentials, suspicious links, or unusual sender domains. Based on these features, the system will classify emails as legitimate, suspicious, or phishing.

Another part of the system will focus on **threat analysis and security evaluation**. This component will examine potential phishing indicators and evaluate the risk level of each email. This helps ensure that the system not only classifies emails but also explains why an email might be dangerous.

When a phishing email is detected, the system will generate an automated alert or recommended response. This may include warning the user, suggesting that the email be reported, or advising further investigation.

Finally, the system will provide a simple interface where users can review flagged emails and see the analysis results. The system will likely be implemented using Python along with machine learning libraries and basic web technologies.

3. Cybersecurity Concepts Covered

This project explores several important cybersecurity concepts related to protecting systems from phishing attacks.

One key concept is social engineering, which is commonly used in phishing attacks. Social engineering relies on manipulating human behavior rather than exploiting technical vulnerabilities. Understanding how attackers use psychological tactics is an important part of identifying phishing attempts.

Another concept involved in this project is threat detection and analysis. The system will identify suspicious indicators such as fake links, domain spoofing, and messages requesting sensitive information. These indicators are often used by attackers to deceive victims.

The project also explores security automation, which is widely used in modern Security Operations Centers (SOCs). Automated systems help security teams detect threats more quickly and respond more efficiently.

In addition, the project involves secure data handling and risk assessment. Email data must be processed carefully to ensure privacy and security. Evaluating the risk level of suspicious emails also helps improve threat detection accuracy.

4. Ethical Issues

There are several ethical considerations involved in developing an automated phishing detection system.

One important issue is the responsible use of artificial intelligence. AI systems can sometimes produce incorrect results if they are not properly designed or tested. For example, the system might incorrectly label a legitimate email as phishing or fail to detect a real phishing message. It is important to evaluate the system carefully and improve accuracy through testing.

Another ethical concern involves privacy and data protection. Email messages may contain personal or sensitive information, so the project must ensure that this data is handled securely. The system will only use email data for analysis within the scope of the project.

False positives and false negatives are also important considerations. If legitimate emails are incorrectly flagged as phishing, it could disrupt communication. On the other hand, if phishing emails are missed, users could still be exposed to attacks. Because of this, the system must balance security with accuracy.

Finally, the project emphasizes ethical cybersecurity practices. The purpose of this system is to improve defensive security and help users recognize phishing threats, not to create tools that could be misused for malicious purposes.

5. Group Member Authorship

Each member of the group contributes to different aspects of the project.

Michael Cruz – Database & Backend Engineer
Responsible for designing the database structure and backend logic used to store email data, classifications, and system logs.

Fatima Saleem – AI System Lead
Responsible for developing the AI and machine learning components used for phishing detection and integrating the model into the system.

Muhammad H. Kardar – Frontend & Development Engineer
Responsible for designing the user interface and ensuring the system can display analysis results clearly.

Imdadul Meraz – Security & Threat Analysis Specialist
Responsible for analyzing phishing attack patterns, defining detection criteria, evaluating system security coverage, and conducting system testing using simulated phishing emails.

Kevin Minchala – Data Engineering Lead
Responsible for collecting and preparing datasets, cleaning data, and supporting the data pipeline used for training and evaluation.